# Shengxiu Sun

Galaxy/AGN observations | AGN - host galaxy co-evolution | Quasar environments | Machine learning

sxsun@stu.pku.edu.cn
Peking University, Beijing
ORCID 0000-0002-1234-552X
github.com/MaxSun-PKU
Google Scholar (56 citations; $h$-index 4)

## Research profile

PhD candidate in astronomy at Peking University studying AGN–host galaxy co-evolution. I develop spectrophotometric decomposition and uncertainty-aware machine-learning methods to separate nuclear and host-galaxy light, then apply them to quasar hosts, changing-look AGNs, near-infrared black-hole mass calibration, and high-redshift AGN populations. I am particularly interested in spatially resolved quasar environments with near-infrared integral-field spectroscopy.

## Education

**PhD candidate in Astronomy**, Kavli Institute for Astronomy and Astrophysics, Peking University — 2022–present
Supervisor: Prof. Linhua Jiang; expected completion 2027.
**Exchange PhD student**, Institute for Theoretical Physics, Heidelberg University — Sep 2025–Sep 2026
**BSc in Astronomy**, School of Astronomy and Space Science, Nanjing University — Sep 2018–Jun 2022

## Research Highlights

**Quasar host galaxies.** Led a joint DESI spectroscopy–Subaru/HSC imaging decomposition of 1,083 extended type-1 quasars at $0.1 < z < 1$. The hosts are predominantly quiescent, with a substantial post-starburst population, establishing a selection-aware baseline for AGN–host studies.

**Changing-look AGNs.** Developed a two-epoch DESI–SDSS–HSC decomposition for 105 changing-look AGNs. In a matched sample of 82 extended objects, post-starburst hosts are more common than in comparison quasars, while the quiescent fractions are not clearly different; the result favours nuclear accretion changes over synchronized galaxy-wide quenching.

**Near-infrared black-hole masses.** Reduced and analysed LBT/LUCI spectra of 27 SDSS-RM quasars and calibrated broad Pa$\alpha$ and Pa$\beta$ single-epoch estimators against reverberation-mapping masses. The intrinsic scatters are 0.28 dex for Pa$\alpha$ and 0.32 dex for Pa$\beta$ within the calibration sample.

**Host stellar kinematics.** Developed a neural network trained on 220,000 forward-modelled AGN spectra to estimate host stellar velocity dispersions, reconstruct host and AGN components, and report calibrated uncertainties. Among 399 low-redshift broad-line AGNs, 159 confidence-selected measurements give an $M_{\rm BH}$–$\sigma_\star$ slope of $4.74 \pm 0.55$ under the adopted uncertainties.

**Publication record.** The selected record comprises four first-author papers or manuscripts and six collaborative papers, including publications in *ApJ* and *Nature Astronomy*; author positions are shown in the full list. Google Scholar reports 56 citations and an $h$-index of 4 (accessed 3 October 2026).

## Research Skills

**Observations** — LBT/LUCI near-infrared spectroscopy; DESI and SDSS spectroscopy; Subaru/HSC imaging; JWST/NIRCam science analysis; NIRSpec IFU science planning; Euclid morphology.

**Methods** — Near-infrared and optical data reduction; PSF-aware image and spectrum decomposition; stellar-population and emission-line modelling; reverberation-mapping and single-epoch mass calibration; mock-data and injection-recovery validation; uncertainty-aware inference.

**Computing** — Python, NumPy, SciPy, Astropy, PyTorch, FSPS, Matplotlib; reproducible HPC workflows.

## Research interests

AGN–host galaxy co-evolution; quasar host galaxies and environments; changing-look AGNs; black-hole mass and host kinematics; spatially resolved near-infrared spectroscopy; Euclid–JWST–LBT science; interpretable machine learning for astronomical spectroscopy.

## Selected Publications and Manuscripts

### First-author papers and manuscript

**Sun, S.** et al. 2026, *ApJ*, 1004, 13, Quiescent Host Galaxies of Extended Quasars Revealed by Spectrophotometric Decomposition, doi.
**Sun, S.** et al. 2026, Changing-look AGNs from DESI. VI. Host Galaxies, arXiv:2607.28735, submitted to ApJ, arXiv.
**Sun, S.** et al. 2026, Broad Paschen Lines as Black Hole Mass Tracers: A Reverberation Mapping Calibration, arXiv:2608.24645, submitted to A&A, arXiv.
**Sun, S.**, Jiang, L., Bosman, S. E. I., & Cheng, Y., Stellar Velocity Dispersions of Quasar Host Galaxies with a Neural Network Trained on Forward-Modelled Spectra, in preparation.

### Collaborative publications (author position shown)

Jiang, D., Jiang, L., **Sun, S.** (3rd author), et al. 2025, *Nature Astronomy*, AGNs Ruled Out as the Dominant Source of Cosmic Reionization, doi.
Pan, Z., Jiang, L., Guo, W.-J., **Sun, S.** (4th author), et al. 2025, *ApJ*, 987, 48, Iron-corrected Single-epoch Black Hole Masses of DESI Quasars at Low Redshift, doi.
Jiang, D., Jiang, L., Fu, S., Zhang, Z., Chen, J., Pan, Z., **Sun, S.** (7th author), et al. 2026, *ApJ*, 1004, 92, COSMOS-3D: Black Hole Mass Estimators and Luminosity Functions of Paschen-line AGNs, doi.
Chen, J., Jiang, L., **Sun, S.** (3rd author), et al. 2025, *ApJ*, Estimating Bolometric Luminosities of Type 1 Quasars with Self-organizing Maps, doi.
Guo, W.-J., Fawcett, V. A., Siudek, M., Li, Y.-R., Cheng, C., Panda, S., Pan, Z., **Sun, S.** (8th author), et al. 2025, *ApJ*, 995, 139, Changing-look AGNs from DESI. IV. Broad-line Evolution, doi.
Yin, G., Jiang, L., Pan, Z., Martini, P., Guo, W.-J., Zou, S., **Sun, S.** (7th author), et al. 2026, *ApJ*, 1008, 60, The Quasar Luminosity Function at $1 < z < 3.5$, doi.

## Teaching

Teaching Assistant, Galaxy Physics, Peking University 2025
Teaching Assistant, Fundamental Astronomy, Peking University 2024
Teaching Assistant, Astronomical Data Processing, Peking University 2024

## Honors and Scholarships

China Scholarship Council (CSC) State-Sponsored Overseas Study Scholarship, Heidelberg University, 2025.
Mercedes-Benz Scholarship, 2022.
Outstanding Research Award, 2022.

## Selected Conferences

**Spotlight talk**, DESI Collaboration Meeting, Durham University, UK 2026
*Exploring AGN host galaxies with DESI*
**Talk**, CSST Annual Science Meeting, Kunming, China 2025
*DESI CL-AGNs viewed in spectrophotometric decomposition*
**Talk**, 27th Guo Shoujing Symposium on Frontiers in Galaxy Cosmology, Changsha, China 2025
*Exploring the properties of quasar host galaxies with spectrophotometric decomposition*
**Talk**, Probing the Genesis of Supermassive Black Holes, Kavli IPMU, Kashiwa, Japan 2024
*Exploring the properties of quasar host galaxies with spectrophotometric decomposition*